NAIVE RIG-FLATNESS IS RIG-FLAT OVER A JACOBSON BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete, with the localization permanence, the rig-closed lifting and the local flatness criterion all written out

Editorial note. The label lemma-rig-flat-naive names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the WAdm^Noeth category, localizations A_{f} and their completions A_{f} bracket, and naive rig-flatness on rig-closed primes
- Jacobson rings and permanence under finite type quotients
- the rig-closed prime lifting lemmas under localization and contraction
- the local criterion for flatness with the generating condition on primes
- finite inclusions of local rings and units

Source reference. Original lemma 00MB: sources/stacks-chow/tags.txt, source lines 817–817. Prerequisite pointer invoked at proof line(s) 3761; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GEB: sources/stacks-chow/tags.txt, source lines 20471–20471. Prerequisite pointer invoked at proof line(s) 3775; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 00GB: sources/stacks-chow/tags.txt, source lines 601–601. Prerequisite pointer invoked at proof line(s) 3765; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGL: sources/stacks-restricted/pinned-original-source.tex, source lines 3650–3650. Prerequisite pointer invoked at proof line(s) 3770; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGC: sources/stacks-restricted/pinned-original-source.tex, source lines 3381–3381. Prerequisite pointer invoked at proof line(s) 3771; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGF: sources/stacks-restricted/pinned-original-source.tex, source lines 3460–3460. Prerequisite pointer invoked at proof line(s) 3789; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGB: sources/stacks-restricted/pinned-original-source.tex, source lines 3333–3333. Prerequisite pointer invoked at proof line(s) 3796; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\varphi : A \to B$ be an arrow of $WAdm^{Noeth}$. If A/I is Jacobson for some (equivalently any) ideal of definition $I \subset A$ and φ is naively rig-flat, then φ is rig-flat.*

Proof. Assume φ is naively rig-flat. We first state some obvious consequences of the assumptions. Namely, let $f \in A$. Then $A, B, A_{\{f\}}, B_{\{f\}}$ are Noetherian adic topological rings. The maps $A \to A_{\{f\}} \to B_{\{f\}}$ and $A \to B \to B_{\{f\}}$ are adic and topologically of finite type. The ring maps $A \to A_{\{f\}}$ and $B \to B_{\{f\}}$ are flat as compositions of $A \to A_f$ and $B \to B_f$ and the completion maps which are flat by Algebra, Lemma 00MB. The quotients of each of the rings $A, B, A_{\{f\}}, B_{\{f\}}$ by I is of finite type over A/I and hence Jacobson too (Algebra, Proposition 00GB).

Let $\mathfrak{q}' \subset B_{\{f\}}$ be rig-closed. It suffices to prove that $(B_{\{f\}})_{\mathfrak{q}'}$ is flat over $A_{\{f\}}$, see Lemma 0GGL. By Lemma 0GGC the primes $\mathfrak{q} \subset B$ and $\mathfrak{p}' \subset A_{\{f\}}$ and $\mathfrak{p} \subset A$ lying under $\mathfrak{q}'$ are rig-closed. We are going to apply Algebra, Lemma 0GEB to the diagram

$$\begin{array}{ccc} B_{\mathfrak{q}} & \longrightarrow & (B_{\{f\}})_{\mathfrak{q}'} \\ \uparrow & & \uparrow \\ A_{\mathfrak{p}} & \longrightarrow & (A_{\{f\}})_{\mathfrak{p}'} \end{array}$$

with $M = B_{\mathfrak{q}}$. The only assumption that hasn't been checked yet is the fact that $\mathfrak{p}$ generates the maximal ideal of $(A_{\{f\}})_{\mathfrak{p}'}$. This follows from Lemma 0GGF; here we use that $\mathfrak{p}$ and $\mathfrak{p}'$ are rig-closed to see that f maps to a unit of $A/\mathfrak{p}$ (this is the only step in the proof that fails without the Jacobson assumption). Namely, this tells us that $A/\mathfrak{p} \to A_{\{f\}}/\mathfrak{p}'$ is a finite inclusion of local rings (Lemma 0GGB) and f maps to a unit in the second one. □